

Ethan Tizabgar

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EDUCATION

University of California, Irvine – Mechanical Engineering

B.S. Mechanical Engineering | Expected 2028

Cumulative GPA: 3.86/4.00

SKILLS

Technical Skills: SolidWorks (CSWA), Arduino, C++

Engineering: CAD Modeling, Technical Drafting, Prototyping, Mechanical Design

Coursework: Engineering Graphics & Design, Statics

WORK EXPERIENCE

SEPTEMBER 2025 - June 2026

Supplemental Instructor (Statics & Engineering Design) — Santa Monica College

- Provide supplemental instruction for statics and engineering design, emphasizing SolidWorks modeling, technical drafting, free-body diagrams, and equilibrium analysis.
- Develop practice problems, review materials, and exam preparation resources in collaboration with the instructor to address common conceptual challenges.
- Lead structured review sessions and provide individualized support to strengthen students' problem-solving skills and troubleshoot SolidWorks modeling errors.

July 2022- January 2025

Swim Instructor — Westside Aquatics

- Instructed children ages 2–13 in swimming, adapting instruction to different developmental levels and learning needs.
- Developed individualized lessons while maintaining a safe learning environment and communicating student progress with parents.

ENGINEERING PROJECTS

May 2026 - June 2026

EMG-Controlled Robotic Hand — Santa Monica College Team Project

- Collaborated with a team to develop an InMoov-inspired tendon-driven robotic hand concept using servo actuation, springs, PTFE tendon routing, and 3D-printed components.
- Integrated a MyoWare 2.0 EMG muscle sensor with an Arduino Uno R4 WiFi to measure forearm muscle activity and translate contractions into gesture commands.
- Developed Arduino-based signal processing, calibration, smoothing, and threshold-based classification to distinguish rock, paper, and scissors gestures.
- Iterated on sensor placement and classification thresholds to improve reliability when different gestures produced similar EMG readings, ultimately demonstrating a functional EMG-based gesture-control system.

January 2025 - February 2025

Wildfire-Response Rainwater System — Santa Monica College Team Project

- Designed a hilltop wildfire-response water tank and developed load-bearing support legs for uneven terrain, accounting for stability, weight distribution, and safety requirements.
- Created and iterated SolidWorks CAD models and prototypes based on testing results, failed iterations, and instructor feedback
- Applied engineering design principles including requirements development, prototyping, design validation, and team-based integration of structural, functional, and environmental requirements.

July 2023 - August 2023

Go-Kart Design & Fabrication — UCLA Engineering Summer Program

- Designed, built, and tested functional go-karts as part of an intensive hands-on mechanical engineering course focused on fundamentals of design, fabrication, and performance evaluation.
- Led a small team through key project phases: concept development, SolidWorks modeling, material selection, and assembly
- Applied problem-solving and troubleshooting skills under pressure to address mechanical failures, optimize drivetrain and frame components, and improve overall vehicle performance.